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Batch - 1 (Full Stack AI)

Assignment - 2

Section A

- i) A symbol table is a data structure used by a compiler to store information about all identifiers in a program. These identifiers include: variables, functions, arrays, objects.

The symbol table plays a crucial role throughout compilation:-

- Stores identifier information.
- Helps in semantic analysis.
- Supports code generation.
- Handles scope management.
- Used in optimization.

Different phases of compiler perform different operations:-

- i) Insertion:-

- Adds new identifier to table.
- Done during lexical/syntax analysis.
- Eg:- `int x;` → insert `x` with `type = int`.

- ii) Lookup (Search):-

- Checks if identifier exists.
- Used in semantic analysis.
- Eg:- `x = 10;` → check if `x` is declared.

- iii) Update

- Modify attributes (type, value, scope).